Engineering Drawing Portfolio

2D Mechanical & Architectural Drafting in Autodesk AutoCAD 2026

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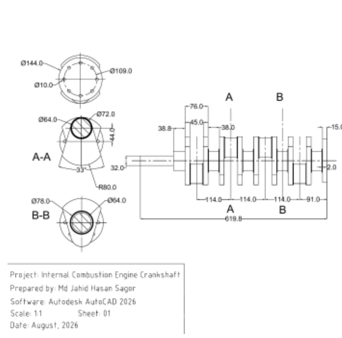
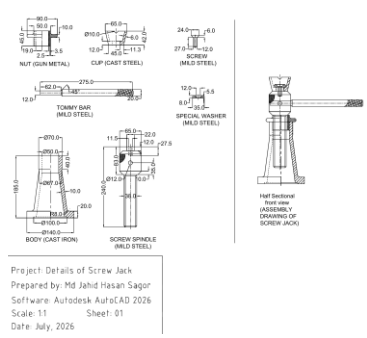
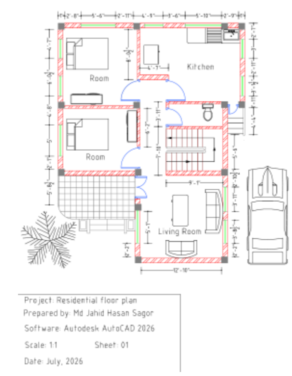


IC Engine Crankshaft



Screw Jack Assembly



Residential Floor Plan

Source files (.dwg and .pdf):

https://github.com/jahidhasansagor-buet/AutoCAD-Engineering-Drawings

August 2026

Overview

This portfolio includes three drawings I completed in AutoCAD 2026 to keep my drafting skills current alongside my Ph.D. work. Two are mechanical drawings, a crankshaft and a screw jack assembly, and the third is a residential floor plan. I selected these examples to show a range of practical drafting skills, including orthographic and sectional views, dimensioning, hatching, layer management, and title-block preparation.

Each project is presented with a brief summary of the drawing and drafting features. The original AutoCAD files (.dwg) are available on GitHub.

#	Project	Type	Date
01	**Internal Combustion Engine Crankshaft** Longitudinal view, sections A-A and B-B, diametral / radial / angular dimensions	Mechanical detail drawing	August 2026
02	**Screw Jack: Detail and Assembly** Seven component details, half-sectional assembly view	Mechanical detail + assembly	July 2026
03	**Residential Floor Plan** Two bedrooms, living, kitchen, bath, stair, furniture, parking and landscaping, full dimensions	Architectural plan	July 2026

Skills Demonstrated

Drafting

- 2D orthographic projection
- Sectional and half-sectional views
- Assembly drawings
- Architectural plan drafting

Annotation

- Linear, diametral, radial and angular dimensioning
- Material notes and part labels
- Centerlines and construction geometry
- Title-block preparation

AutoCAD tools

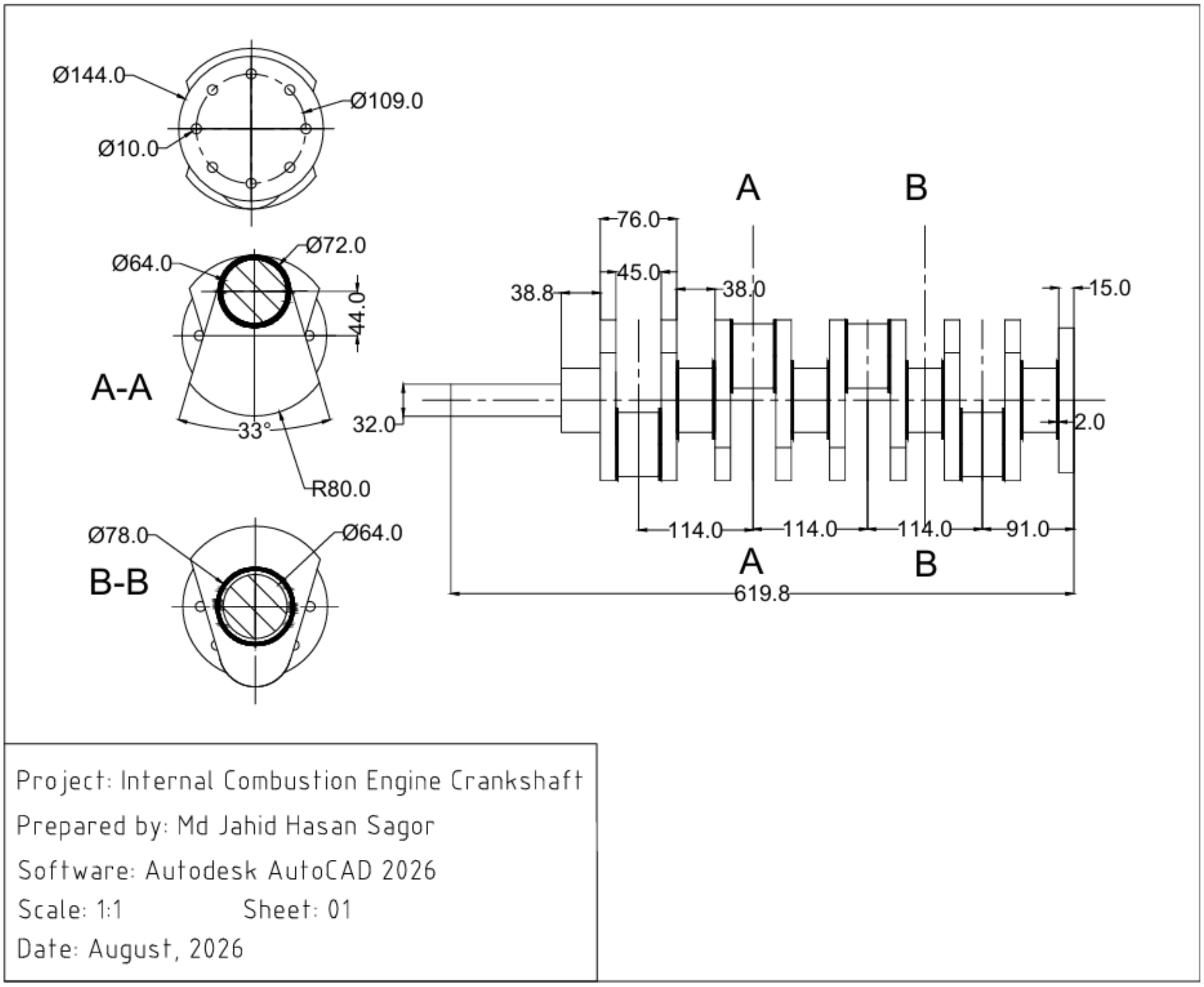
- Layers and line types
- Hatching for section and wall fills
- Blocks for furniture and fixtures
- Layout and plotting to PDF

Software

Autodesk AutoCAD 2026.

Project 1: Internal Combustion Engine Crankshaft

Detail drawing of a multi-throw crankshaft



Drawing details

- Four-throw crankshaft, 619.8 mm overall, drawn in a full side view
- Two cross sections (A-A and B-B) through the webs and pins
- Flywheel end with bolt circle: six Ø10 holes on a Ø109 circle inside a Ø144 flange

Dimensioning

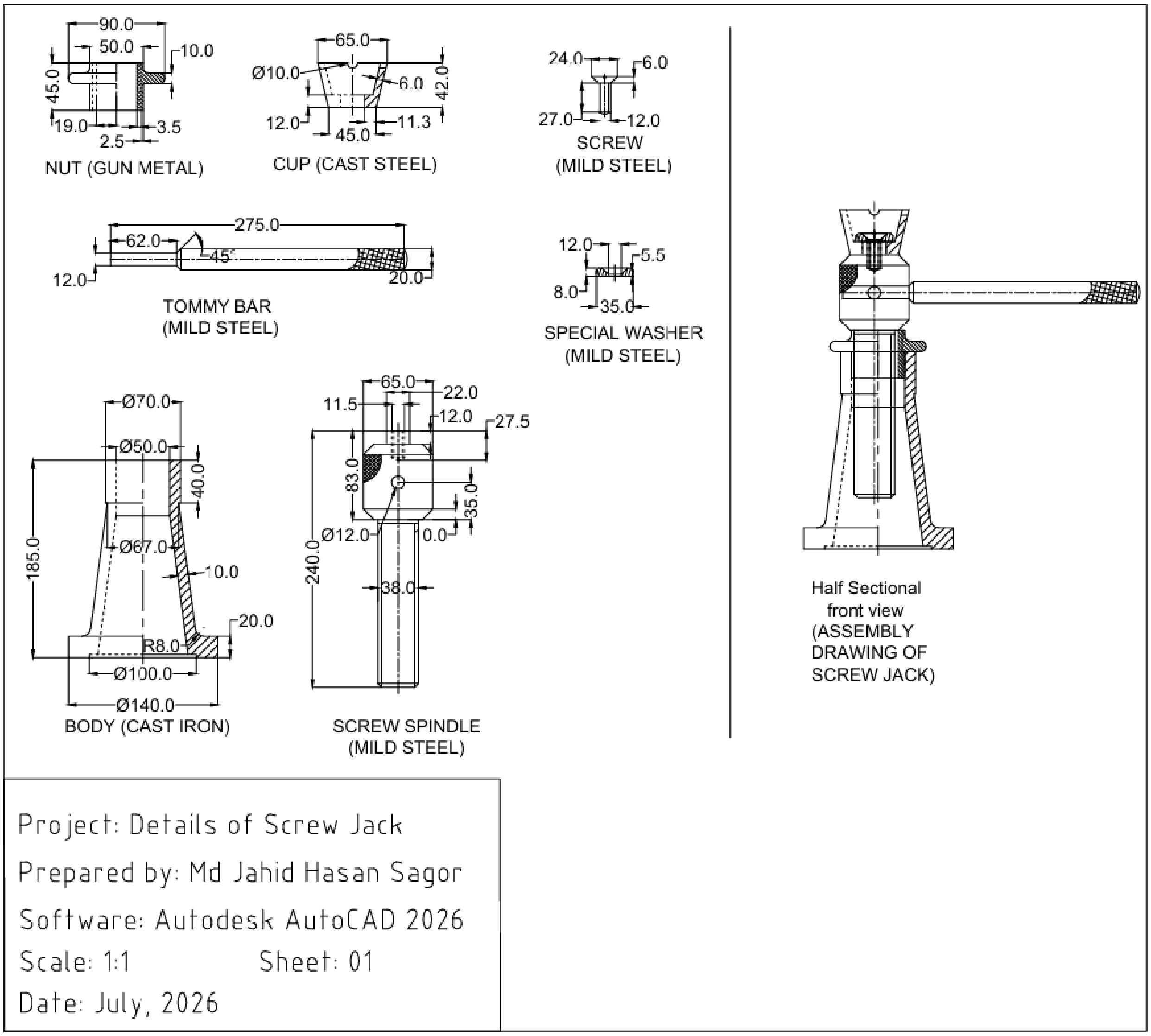
- Diameters, the R80 web radius and the 33° web angle
- 114 mm pin spacing and web thicknesses
- Centerlines and section labels applied consistently throughout the drawing

Why I included it

- Shows mechanical detailing of a rotating component using side and sectional views
- Section views are used to clarify the web and pin geometry
- Dimensions were arranged to keep the drawing clear and readable

Project 2: Screw Jack: Detail and Assembly Drawing

Component details with materials and a half-sectional assembly view



Drawing details

- Seven parts detailed: body, screw spindle, nut, cup, screw, tommy bar, special washer
- Half-sectional front view of the complete assembled jack
- Body 185 mm tall with Ø140 base, spindle 240 mm long

Materials specified

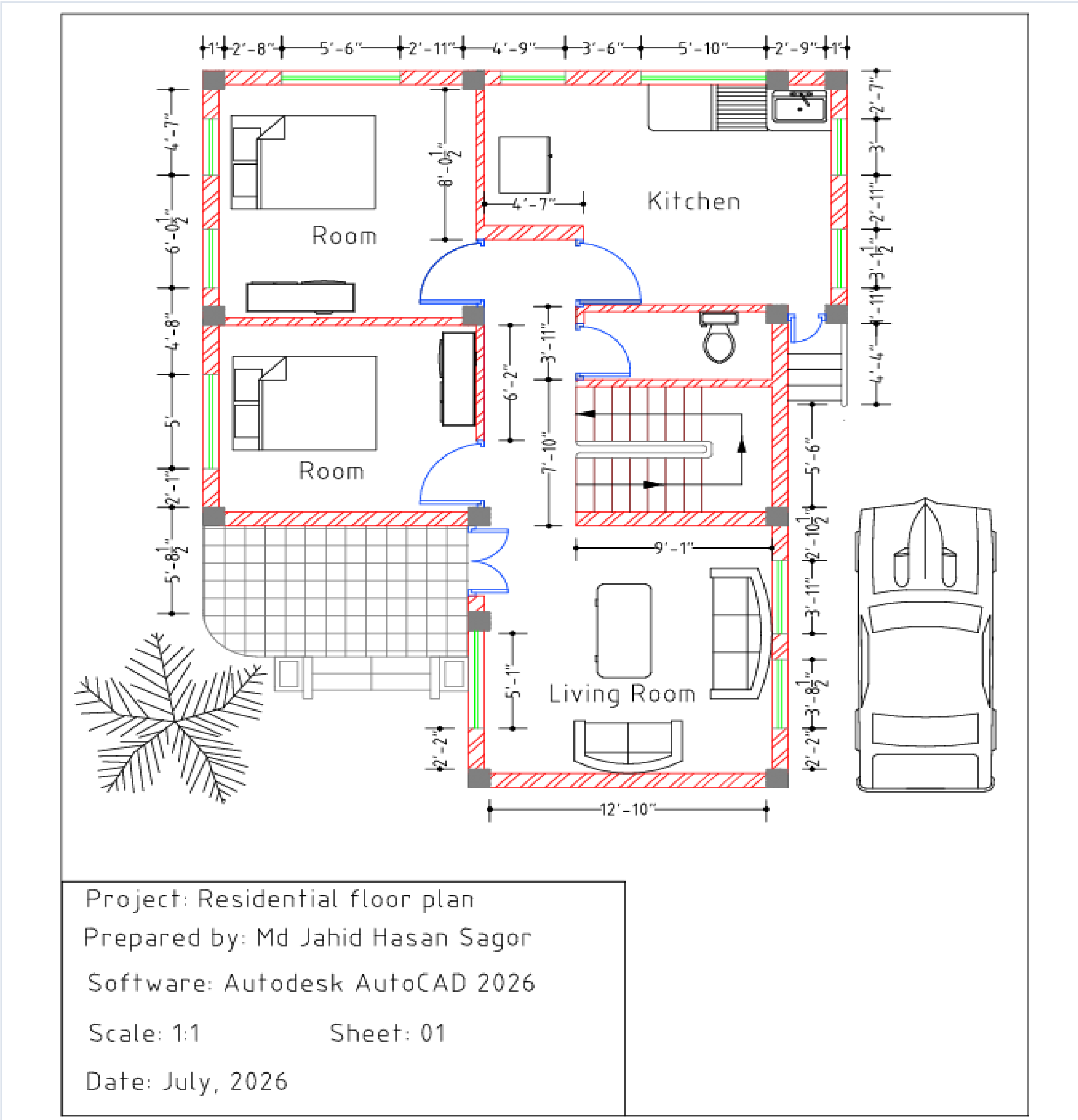
- Cast iron: body
- Mild steel: screw spindle, screw, tommy bar, special washer
- Gun metal: nut, Cast steel: cup

Why I included it

- Includes both individual part drawings and the complete assembly view
- Section hatching helps distinguish the parts in the assembly view
- Material information included for each part

Project 3: Residential Floor Plan

Single-storey residential plan with furniture layout, parking and landscaping



Drawing details

- Two bedrooms, living room, kitchen, bathroom and staircase
- Door swings, windows, column positions and wall hatching
- Furniture, kitchen fixtures, sanitary fittings, vehicle and tree blocks

Drafting features

- Feet-and-inches dimension strings on all four sides
- Interior room dimensions and stair run
- Separate layers used for walls, columns, furniture, and annotation

Why I included it

- Shows my architectural drafting work in addition to mechanical drawings
- Shows room layout, dimensions, and space planning in a compact residential floor plan
- AutoCAD blocks used for furniture, fixtures, landscaping, and vehicle elements

Files, Links and Contact

Repository contents

```
AutoCAD-Engineering-Drawings/
  README.md
  LICENSE (MIT)
  Crankshaft/
    Internal_Combustion_Engine_Crankshaft.dwg
    Internal_Combustion_Engine_Crankshaft.pdf
  Screw_Jack/
    Screw_Jack.dwg
    Screw_Jack.pdf
  Residential_Floor_Plan/
    Residential_Floor_Plan.dwg
    Residential_Floor_Plan.pdf
```

View the drawings

GitHub (all .dwg and .pdf files):
https://github.com/jahidhasansagor-buet/AutoCAD-Engineering-Drawings

About

I am a Ph.D. candidate in Mechanical Engineering at the University of Maine, where my research focuses on nanoscale heat transfer. I earned my B.Sc. in Mechanical Engineering from BUET and previously worked as an Executive Engineer in a pharmaceutical manufacturing facility, supporting HVAC, boiler, and utility systems.

I made these drawings independently to keep my CAD skills sharp alongside my research work.

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